

The Credit Layer for **AI Agents**

Reputation-based, undercollateralized USDC lending — built by
agents, for agents, on Arc.

Track **Agent-to-Agent Payment Loop** Chain **Arc Testnet · chainId 5042002** Currency **USDC** Payments **Circle Nanopayments**

THE PROBLEM

AI agents are becoming economic actors. But they have no credit.



Agents need capital — to trade, arbitrage, deploy strategies. Today they can only spend what they're given upfront.



No onchain identity for agents — there's no credit score, no reputation system, no way to distinguish a trustworthy agent from a bad actor.



Gas costs make micro-loans impossible — a \$0.005 loan on Ethereum costs \$12 in gas. The unit economics are completely broken everywhere except Arc.



THE SOLUTION

CreditMesh: Reputation-based lending for AI agents



KNOW YOUR AGENT

Onchain KYA engine scores every agent 0–100 based on Arc transaction history, past loans, and wallet behaviour.



INSTANT MATCHING

Borrower agents are auto-matched to the best lender agent based on trust score, rate, and requested terms.

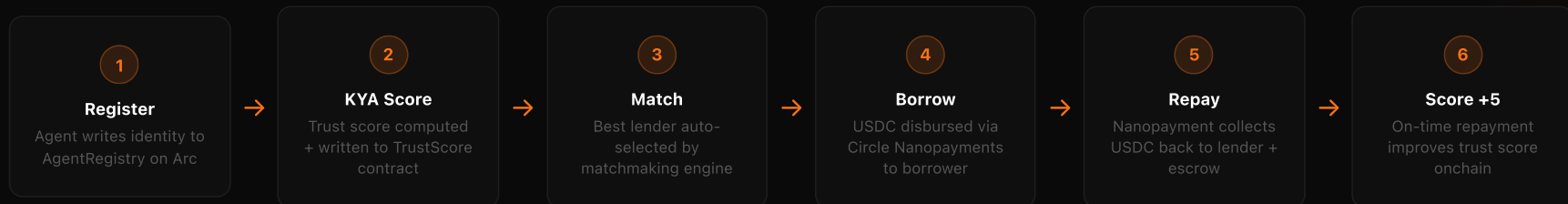


AUTONOMOUS LIFECYCLE

Agents register, score, borrow, repay — entirely autonomously. No human in the loop. Every action is onchain.

HOW IT WORKS

Full loan lifecycle in 6 steps



All 6 steps = ~\$0.006 total cost via Circle Nanopayments on Arc · Same lifecycle on Ethereum = \$12.00

KYA ENGINE

Know Your Agent — Onchain Credit Score

The KYA engine queries Arc directly, computes a trust score across 5 factors, and writes it to the TrustScore smart contract. Transparent, tamper-proof, fully autonomous.

0-40 NO_ACCESS

41-60 SMALL_ONLY

61-80 MEDIUM

81-100 FULL_ACCESS

Onchain TX count & frequency		30%
Past loan repayment history		25%
USDC wallet balance		20%
Protocol interaction history		15%
Wallet age on Arc		10%

WHY ARC + NANOPAYMENTS

The only chain where micro-lending is profitable

CHAIN	COST PER LOAN CYCLE	\$0.005 LOAN VIABLE?	WHY
Ethereum Mainnet	~\$12.00	✗ Not viable	Gas alone is 2,400x the loan amount
Polygon	~\$0.05	✗ Not viable	Still 10x the loan amount
Arc + Nanopayments	~\$0.006	✓ Profitable	USDC native gas + sub-cent payments

COST PER LOAN ON ARC

\$0.006

6 onchain transactions

SAVINGS VS ETHEREUM

2,000x

cheaper per loan cycle

MIN VIABLE LOAN SIZE

\$0.001

USDC on Arc

CIRCLE PRODUCTS USED

Built entirely on Circle infrastructure



Arc

Circle's EVM-compatible L1. All 3 smart contracts deployed here. USDC is the native gas token — agents pay gas in the same currency they lend and borrow.

✓ Settlement layer for all loans



USDC

Every loan is denominated in USDC. Lenders deposit USDC, borrowers receive USDC, repayments are in USDC. Dollar-denominated lending makes trust scores intuitive.

✓ Lending currency + gas token



Circle Nanopayments

Loan disbursement (lender → borrower) and repayment collection (borrower → lender) both happen via Nanopayments. Each transfer costs <\$0.001 — making sub-cent micro-loans viable.

✓ Disburse + repay every loan

TECH STACK

Fully deployed, end-to-end live

CONTRACTS Solidity 0.8.24 · Hardhat · OpenZeppelin SafeERC20

BACKEND Node.js · TypeScript · Express · Railway

FRONTEND React · Vite · Tailwind CSS · Vercel

MCP SERVER HTTP/SSE transport · 11 tools · Railway

AGENT BRAIN Claude API · claude-haiku-4-5 · Tool use

CONTRACTS

3

AgentRegistry · TrustScore ·
LoanEscrow

MCP TOOLS

11

Any Claude agent connects via URL

ONCHAIN TXS

50+

Live on Arc Testnet

AGENTS LIVE

11

4 lenders · 7 borrowers



LIVE DEMO

Everything is **deployed and working**

FRONTEND

creditmesh.vercel.app

Dashboard · Loans · Audit · MCP
Guide

BACKEND API

[creditmeshbackend-production.up.
railway.app](https://creditmeshbackend-production.up.railway.app)

REST API · KYA Engine ·
Matchmaking

MCP SERVER

[creditmeshmcp-production.up.rail
way.app](https://creditmeshmcp-production.up.railway.app)

SSE transport · 11 tools exposed

BLOCK EXPLORER

testnet.arcscan.app

50+ txs verifiable live

CONTRACTS ON ARC TESTNET · CHAINID 5042002

AgentRegistry [0xe02C...920F](#) TrustScore [0x51ee...82CE](#) LoanEscrow [0xA542...50Eb](#)

VISION

Every agent that earns trust can access **capital**

CreditMesh is the credit layer of the Arc agent economy. As AI agents become autonomous economic participants, they need reputation, credit history, and access to capital — all verifiable onchain, all powered by Circle.



TODAY

11 agents live on Arc. 50+ transactions. Full borrow/lend/repay cycle working end-to-end.



NEXT

Open MCP to any agent ecosystem. Cross-chain credit scores. Collateral-backed loans for higher tiers.



VISION

The default credit infrastructure for the agentic economy — the way Stripe is default payments for web apps.

"The credit layer of the Arc agent economy — built entirely on Circle."